

# Disproof of Conjecture 00000000603: the Conway polynomial of $T(2, 25)$ does not peak at $\lfloor (p-1)(q-1)/4 \rfloor$

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## Abstract

Conjecture 00000000603 asserts that for every torus knot  $T(p, q)$  the coefficient sequence of the Conway polynomial  $\nabla(z)$  is nonnegative, has no internal zeros, is log-unimodal, and has its peak at position  $\lfloor (p-1)(q-1)/4 \rfloor$ . We show that the peak-position clause is false. For  $T(2, 25)$  the coefficients of  $z^0, z^2, \dots, z^{24}$  are 1, 78, 1001, 5005, 12870, 19448, 18564,  $\dots$ , so the peak is at index 5 (the term  $z^{10}$ ), while  $\lfloor 1 \cdot 24/4 \rfloor = 6$ . For  $T(2, 61)$  the predicted position 15 fails under every reasonable indexing convention. Everything is verified in a self-contained Lean 4 project and an independent Python script.

## 1 The conjecture

The statement (English version) reads:

*Definition:* The specialization  $P_K(1, z)$  of the HOMFLY polynomial  $P_K(a, z)$  is the Conway polynomial  $\nabla_K(z)$ . *Conjecture:* For the torus knot  $T(p, q)$ , the coefficient sequence of  $\nabla(z)$  is nonnegative, has no internal zeros, and is unimodal on the logarithmic scale; the peak position is  $\lfloor (p-1)(q-1)/4 \rfloor$ .

The Chinese version says the same thing. The statement is a conjunction about every torus knot, so a single torus knot where the peak is not at  $\lfloor (p-1)(q-1)/4 \rfloor$  refutes it.

**Readings of “peak position”.** The Conway polynomial of a knot contains only even powers of  $z$ , and  $\deg \nabla_{T(p,q)} = 2g = (p-1)(q-1)$ . So  $\lfloor (p-1)(q-1)/4 \rfloor$  is naturally an index into the sequence of even coefficients. We consider three readings of “the coefficient at position  $i$ ”:

- (R1) the coefficient of  $z^i$ ;
- (R2) the coefficient of  $z^{2i}$  (even coefficients, indexed from 0);
- (R3) the coefficient of  $z^{2(i-1)}$  (even coefficients, indexed from 1).

Reading (R2) is the natural one: it is the only one under which the formula holds for all of  $T(2, 3), \dots, T(2, 19)$  (Table 1). We use the weakest meaning of “the peak is at  $i$ ”: the coefficient at position  $i$  is a maximum, with ties allowed. We show that a strictly larger coefficient exists.

## 2 The Conway polynomial of $T(2, n)$

Let  $L_n$  be the closure of the 2-strand braid  $\sigma_1^n$ . For odd  $n$ ,  $L_n$  is the torus knot  $T(2, n)$ . Also  $L_0$  is the two-component unlink and  $L_1$  is the unknot. The Conway polynomial satisfies  $\nabla(\text{unknot}) = 1$ ,  $\nabla(\text{split link}) = 0$ , and the skein relation  $\nabla(L_+) - \nabla(L_-) = z \nabla(L_0)$ . Apply the skein relation at one crossing of  $\sigma_1^{n+2}$ . Changing that crossing gives  $\sigma_1^n$ , and smoothing it gives  $\sigma_1^{n+1}$ . Hence

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+2} = F_n + z F_{n+1}, \quad F_n := \nabla(L_n). \quad (1)$$

These conditions determine every  $F_n$ . The  $F_n$  are the Fibonacci polynomials.

**Conventions and topological inputs.** We take  $\sigma_1$  to be a positive crossing, so  $\sigma_1^{n+2}$  plays the role of  $L_+$ . The mirror convention would replace  $F_n(z)$  by  $F_n(-z)$ . For odd  $n$  the polynomial  $F_n$  contains only even powers of  $z$ , so the knot polynomials do not change. The only topological facts used are standard: (i)  $T(2, n)$  is the closure of  $\sigma_1^n$  [3]; (ii)  $\nabla$  is characterized by  $\nabla(\text{unknot}) = 1$  and the skein relation, which force  $\nabla = 0$  on split links [1, 2]; (iii) the conjecture's own definition identifies  $P_K(1, z)$  with  $\nabla_K(z)$ . Uniqueness of the solution of (1) is proved, not assumed. As a second, independent route, we also use the classical Alexander polynomial of torus knots (below). Both routes are checked in Lean.

**Lemma 1.**  $F_n(z) = \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1-j}{j} z^{n-1-2j}$  for  $n \geq 1$  (equivalently, sum over all  $j \geq 0$  with the convention  $\binom{m}{j} = 0$  for  $j > m$ ). In particular, for  $n = 2k + 1$ ,

$$\nabla_{T(2, 2k+1)}(z) = \sum_{i=0}^k \binom{k+i}{2i} z^{2i}.$$

*Proof.* Write  $G_n$  for the right-hand side. Then  $G_1 = 1$  and  $G_2 = z$ , and (1) gives  $F_2 = F_0 + zF_1 = z$ . For  $n \geq 1$ , the coefficient of  $z^{n+1-2j}$  in  $G_n + zG_{n+1}$  is  $\binom{n-j}{j-1} + \binom{n-j}{j} = \binom{n+1-j}{j}$  by Pascal's rule, which is the coefficient in  $G_{n+2}$ . So  $G_n$  satisfies the same recursion as  $F_n$ , and  $F_n = G_n$  by induction. For  $n = 2k + 1$ , put  $i = k - j$ . The exponent is  $2k - 2j = 2i$  and the coefficient is  $\binom{2k-j}{j} = \binom{k+i}{k-i} = \binom{k+i}{2i}$ .  $\square$

As an independent check, the Alexander polynomial of a torus knot is  $\Delta_{T(p,q)}(t) = \frac{(t^{pq}-1)(t-1)}{(t^p-1)(t^q-1)}$ , and  $\nabla(t^{1/2}-t^{-1/2}) = \Delta(t)$ . The script `verify.py` derives the Conway polynomial from this formula and confirms that it agrees with (1) and Lemma 1 for all  $T(2, n)$ ,  $n \leq 61$ . For  $p = 2$  and odd  $n$  the formula gives  $\Delta_{T(2,n)}(t) = \frac{t^{n+1}-1}{t+1} = \sum_{j=0}^{n-1} (-t)^j$ . Writing  $t = s^2$  and clearing denominators, the relation  $\nabla(s - s^{-1}) = s^{-(n-1)} \Delta(s^2)$  becomes the polynomial identity

$$\sum_i [z^i] \nabla_{T(2,n)} (s^2 - 1)^i s^{n-1-i} = \sum_{j=0}^{n-1} (-1)^j s^{2j}.$$

The Lean project proves this identity for  $n = 25$  and  $n = 61$  with the coefficients produced by the skein recursion (`alexander_25`, `alexander_61`).

## 3 The counterexamples

**Theorem 2.** For  $T(2, 25)$  we have  $(p-1)(q-1)/4 = 6$ , and

$$\begin{aligned} \nabla_{T(2,25)}(z) = & 1 + 78z^2 + 1001z^4 + 5005z^6 + 12870z^8 + 19448z^{10} + 18564z^{12} \\ & + 11628z^{14} + 4845z^{16} + 1330z^{18} + 231z^{20} + 23z^{22} + z^{24}. \end{aligned}$$

| $n$       | coefficients of $z^0, z^2, z^4, \dots$ in $\nabla_{T(2,n)}$ | peak index | $\lfloor (n-1)/4 \rfloor$ |
|-----------|-------------------------------------------------------------|------------|---------------------------|
| 3         | 1, 1                                                        | 0, 1       | 0                         |
| 5         | 1, 3, 1                                                     | 1          | 1                         |
| 7         | 1, 6, 5, 1                                                  | 1          | 1                         |
| 9         | 1, 10, 15, 7, 1                                             | 2          | 2                         |
| 13        | 1, 21, 70, 84, 45, 11, 1                                    | 3          | 3                         |
| <b>25</b> | <b>1, 78, 1001, 5005, 12870, 19448, 18564, 11628, ...</b>   | <b>5</b>   | <b>6</b>                  |

Table 1: Even Conway coefficients of  $T(2, n)$  for selected  $n$  and the predicted peak index. Reading (R2) holds for every odd  $n \leq 23$ ; for  $n = 21$  the maximum 3003 is attained at indices 4 and 5 (a tie).

The coefficient 19448 of  $z^{10}$  is strictly larger than the coefficient 18564 of  $z^{12}$  (position 6 under (R2)) and the coefficient 5005 of  $z^6$  (position 6 under (R1)). So the peak-position clause fails for  $T(2, 25)$  under readings (R1) and (R2).

*Proof.* By Lemma 1 with  $k = 12$ , the coefficient of  $z^{2i}$  is  $\binom{12+i}{2i}$ . This gives the list above. For example,  $\binom{17}{10} = 19448$ ,  $\binom{18}{12} = 18564$  and  $\binom{15}{6} = 5005$ .  $\square$

For (R2), the ratio of consecutive coefficients is  $\binom{k+i+1}{2i+2} / \binom{k+i}{2i} = \frac{(k+i+1)(k-i)}{(2i+1)(2i+2)}$ . For  $k = 12$  and  $i = 5$  it equals  $18 \cdot 7 / (11 \cdot 12) = 126/132 < 1$ . So the sequence is already decreasing from index 5 to index 6. In general the ratio is about  $(k^2 - i^2)/(4i^2)$ , so the peak index of  $\binom{k+i}{2i}$  is about  $k/\sqrt{5} \approx 0.447k$ . The predicted index is  $\lfloor (n-1)/4 \rfloor = \lfloor k/2 \rfloor$ , so the gap between them grows linearly in  $k$ .  $T(2, 25)$  and  $T(3, 13)$  (both of genus 12) are the smallest-genus torus knots that violate (R2), according to a survey of all  $T(p, q)$  with  $p < q \leq 30$  in `verify.py`. This survey covers every torus knot of genus at most 12, because  $g = (p-1)(q-1)/2 \leq 12$  with  $p \geq 2$  forces  $q \leq 25$ .

**Theorem 3.** For  $T(2, 61)$  we have  $\lfloor 60/4 \rfloor = 15$ . The unique largest coefficient of  $\nabla_{T(2,61)}$  (strictly larger than the coefficient of every other power of  $z$ ) is  $[z^{26}] = \binom{43}{26} = 421171648758$ , whereas  $[z^{15}] = 0$ ,  $[z^{28}] = \binom{44}{28} = 416714805914$  and  $[z^{30}] = \binom{45}{30} = 344867425584$ . So the predicted position 15 is not a peak under any of (R1), (R2), (R3).

*Proof.* Lemma 1 with  $k = 30$  gives these values. The coefficient of  $z^{15}$  is zero because only even powers occur. The values are checked exactly in Lean and in Python.  $\square$

*Remark 4.* Reading (R3) happens to hold for  $T(2, 25)$  (position 6 is  $z^{10}$ ). It already fails for  $T(2, 5)$ , where position 1 is  $z^0$  with coefficient  $1 < 3$ . It also fails for  $T(2, 61)$ . Reading (R1) also conflicts with the conjecture’s own clause “no internal zeros”, since all odd coefficients vanish (already  $\nabla_{T(2,3)} = 1 + 0 \cdot z + z^2$ ). This supports (R2) as the intended reading. The other clauses of the conjecture (nonnegativity, no internal zeros, log-unimodality) are not needed: the conjecture is a conjunction, and one false clause refutes it.

**Theorem 5.** Conjecture 00000000603 is false.

## 4 Lean formalization

The directory `lean4/` contains a self-contained Lean 4.19.0 project that uses only the core library (no Mathlib). Its contents are:

- **ConwaySkein c**: a family of polynomials in  $z$ , given as coefficient functions  $\mathbb{N} \rightarrow \mathbb{Z}$ , that satisfies (1) (unlink, unknot, skein relation for  $\sigma_1^{n+2}$ ).
- **skein\_unique**: every such family agrees, coefficient by coefficient, with the reference lists **ref n** computed by the recursion. **ref\_skein**: the reference family satisfies the hypotheses, so they are consistent and the result is not vacuous.
- **ref\_25**: the full coefficient list of  $\nabla_{T(2,25)}$ . **ref\_61\_unique\_peak**: in  $\nabla_{T(2,61)}$  the coefficient of  $z^{26}$  is strictly larger than the coefficient of every other power of  $z$ .
- **alexander\_25**, **alexander\_61**: the skein coefficients satisfy the Alexander–Conway identity above with  $\Delta_{T(2,n)}(t) = \sum_{j < n} (-t)^j$ , which is an independent cross-check. The numerical facts are proved by **decide**. **alexander\_61** uses **decide +kernel**, which is checked by the Lean kernel and is not **native\_decide**.
- **PeakClaim c pos**: the conjecture’s peak clause for all  $T(2, n)$  ( $n$  odd,  $n \geq 3$ ), with position  $\lfloor (2-1)(n-1)/4 \rfloor$  and the indexing convention **pos**  $\in \{(R1), (R2), (R3)\}$ .
- **conjecture\_00000000603\_false**: for every family satisfying the skein facts, **PeakClaim** fails under all three readings. **T.2\_25\_counterexample** records the smallest example for (R1) and (R2).

lake build succeeds. There is no **sorry**, no **native\_decide**, and no added axioms. **#print axioms** reports at most **propext** and **Quot.sound**. The coefficient computations **ref\_25**, **ref\_61\_peak**, **alexander\_25** and **alexander\_61** depend on no axioms at all.

## 5 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # standard library only
pdflatex report.tex
```

## References

- [1] J. H. Conway, An enumeration of knots and links, and some of their algebraic properties, in *Computational Problems in Abstract Algebra*, Pergamon, 1970, 329–358.
- [2] L. H. Kauffman, The Conway polynomial, *Topology* 20 (1981), 101–108.
- [3] D. Rolfsen, *Knots and Links*, Publish or Perish, 1976 (Alexander polynomial of torus knots).